

NISHANT KUMAR

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Education

Indian Institute of Technology, Goa <i>B.Tech, Mathematics and Computing Engineering</i>	2019 – 2023
Jawahar Navodaya Vidyalaya, Bangalore Urban <i>Class 12, CBSE</i>	2016 – 2018

Experience

Senior Member of Technical Staff <i>Siemens EDA, Noida</i>	Jan 2024 – Present
- Optimizing HDL Editing workflows within Tessent DFT toolchain, focusing on high-performance RTL parsing and elaboration for massive-scale designs. - Resolved 50+ critical bugs involving HIE (Hierarchical Instance Expansion), structs, and complex signal connectivity in Verilog/VHDL. - Automated RTL view validation across multiple synthesis streams, reducing manual verification time by 20%. - Improved C++ memory efficiency and execution speed in HDLE modules by refactoring data structures and algorithms. - Enhanced critical elaboration stages including type resolution, width computation, and assign de-compilation, ensuring 100% tool accuracy.	
Data Scientist <i>Turtlemint, Pune</i>	Jul 2023 – Jan 2024
- Engineered end-to-end machine learning pipelines in Python, improving data cleaning speed by 15% via modular preprocessing scripts. - Segmented 10,000+ POSPs (Point of Sales Persons) using K-means clustering, enabling targeted advertising that lowered acquisition costs. - Processed and analyzed large-scale datasets using PySpark and AWS (S3/EC2), facilitating data-driven decision making. - Conducted A/B testing on marketing models to validate feature importance and model scalability.	
Summer Intern <i>Siemens EDA, Noida</i>	Jun 2022 – Dec 2022
- Scaled debug infrastructure by developing TCL scripts for automated validation of modification commands. - Leveraged GDB and Linux tools to perform deep-dive debugging of Object-Oriented C++ codebases. - Documented system architecture and test coverage, identifying edge cases in netlist modification features.	

Projects

AdvHouse PredApp — Python, Flask, Scikit-Learn	Jan 2022 – Mar 2022
- Built a house price prediction engine with 89% accuracy using Ridge and Lasso Regression to prevent overfitting. - Performed feature engineering on the Ames Housing dataset, handling 80+ variables to extract key drivers of valuation. - Deployed a responsive Flask web app on Heroku, integrating a user-friendly UI for real-time price estimation.	

COVID-19 Forecasting — Mathematical Modeling

Apr 2020 – May 2020

- Developed a modified **SIR (Susceptible-Infected-Recovered)** model to simulate pandemic spread in Punjab, India.
- Achieved a high correlation between model projections and actual case data by tuning infection and recovery rate parameters.
- Visualized trends using Matplotlib to communicate peak infection windows to non-technical stakeholders.

Skills

Languages: C, C++, Python, Verilog, VHDL, SQL, JavaScript, HTML/CSS, TCL

Frameworks & Tools: React.js, Flask, PySpark, AWS, AutoCAD, Linux, GDB, Git, LaTeX

Core Competencies: Data Structures (DSA), Object-Oriented Programming (OOP), Machine Learning, RTL Synthesis, DFT, RTL Design, Algorithm Design

Soft Skills: Technical Problem Solving, Cross-functional Collaboration, Mentorship

Positions of Responsibility

- **Student Mentor:** Facilitated technical workshops and peer-mentoring for juniors at IIT Goa, improving student performance in core CS courses.
- **Volunteer:** Managed logistics and proctoring for national-level talent searches (NDST/JDST) for Dakshana Foundation.